Python Engineering - Class 01

From Equation to Algorithm | EX01 - Dynamic Pressure

Class objective: Learn to translate an engineering problem from an equation on paper into a sequence of executable Python instructions.

This document is the base exercise guide. It is not a full solution to copy: it organizes the problem, implementation steps, reference values and required validation.

Path

```
01_python_bootcamp/
`-- week_01_logic/class_01_equation_to_algorithm/
    |-- notes/
    `-- exercises/
```

Required workflow

```
PROBLEM
INPUTS
OUTPUTS
EQUATIONS
SPECIAL CASES
DO I NEED ITERATION?
ALGORITHM
VALIDATION
```

Before coding, write these eight sections. The goal is to separate problem reasoning from syntax.

EX01 - Dynamic Pressure

Problem: Calculate dynamic pressure q from air density rho and speed magnitude.

Reference inputs: rho = 1.225 kg/m^3; speed = 100.0 m/s

Equations / relationship: q = 0.5 * rho * speed**2

Expected check: Expected q = 6125 Pa. Validate rho >= 0 and speed >= 0.

Implementation guide

- 1. Identify inputs and output.
- 2. Translate the equation into Python operators.
- 3. Validate physically impossible values.
- 4. Compute q.
- 5. Display the result with units.

Before running

Predict sign, order of magnitude, data type and any relevant shape. Then compare the prediction with the actual output.

Final validation

Answer these questions as comments at the end of the exercise. The goal is not memorized wording but demonstrating that you can explain the code flow.

- Explain the difference between an equation and an algorithm.
- What is the difference between int and float?
- What does ** do?
- Why is speed=-100 invalid if speed is a magnitude?
- Manually verify q for rho=1.225 and speed=100.

Completion criterion

The class is complete when the code runs, the reference checks make sense, you can explain every block without reading a solution, and you can modify at least one input while predicting the expected effect.